

Configuring NVMe/TCP and Multipathing for RHEL and SLES Linux OS with VSP One Block 20 Storage NVM Subsystems

v1.0

Implementation Guide

This document shows how to configure NVMe/TCP on Hitachi Virtual Storage Platform One Block 20 (VSP One B20) family storage systems with RHEL and SLES Linux operating systems. This document applies to VSP One B24, B26, and B28 storage systems.

Hitachi Vantara

July 2024

© 2024 Hitachi Vantara LLC. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying and recording, or stored in a database or retrieval system for commercial purposes without the express written permission of Hitachi, Ltd., or Hitachi Vantara Corporation (collectively, "Hitachi"). Licensee may make copies of the Materials provided that any such copy is: (i) created as an essential step in utilization of the Software as licensed and is used in no other manner; or (ii) used for archival purposes. Licensee may not make any other copies of the Materials. "Materials" mean text, data, photographs, graphics, audio, video and documents.

Hitachi reserves the right to make changes to this Material at any time without notice and assumes no responsibility for its use. The Materials contain the most current information available at the time of publication.

Some of the features described in the Materials might not be currently available. Refer to the most recent product announcement for information about feature and product availability, or contact Hitachi Vantara Corporation at https://support.HitachiVantara.com/en_us/contact-us.html.

Notice: Hitachi products and services can be ordered only under the terms and conditions of the applicable Hitachi agreements. The use of Hitachi products is governed by the terms of your agreements with Hitachi Vantara Corporation.

By using this software, you agree that you are responsible for:

- 1) Acquiring the relevant consents as may be required under local privacy laws or otherwise from authorized employees and other individuals to access relevant data; and
- 2) Verifying that data continues to be held, retrieved, deleted, or otherwise processed in accordance with relevant laws.

Notice on Export Controls. The technical data and technology inherent in this Document may be subject to U.S. export control laws, including the U.S. Export Administration Act and its associated regulations, and may be subject to export or import regulations in other countries. Reader agrees to comply strictly with all such regulations and acknowledges that Reader has the responsibility to obtain licenses to export, re-export, or import the Document and any Compliant Products.

EXPORT CONTROLS - Licensee will comply fully with all applicable export laws and regulations of the United States and other countries, and Licensee shall not export, or allow the export or re-export of, the Software, API, or Materials in violation of any such laws or regulations. By downloading or using the Software, API, or Materials, Licensee agrees to the foregoing and represents and warrants that Licensee is not located in, under the control of, or a national or resident of any embargoed or restricted country.

Hitachi is a registered trademark of Hitachi, Ltd., in the United States and other countries.

AIX, AS/400e, DB2, Domino, DS6000, DS8000, Enterprise Storage Server, eServer, FICON, Flash Copy, IBM, Lotus, MVS, OS/390, PowerPC, RS6000, S/390, System z9, System z10, Tivoli, z/OS, z9, z10, z13, z/VM, BCPI™ and z/VSE are registered trademarks or trademarks of International Business Machines Corporation.

Active Directory, ActiveX, Bing, Excel, Hyper-V, Internet Explorer, the Internet Explorer logo, Microsoft, the Microsoft Corporate Logo, MS-DOS, Outlook, PowerPoint, SharePoint, Silverlight, SmartScreen, SQL Server, Visual Basic, Visual C++, Visual Studio, Windows, the Windows logo, Windows Azure, Windows PowerShell, Windows Server, the Windows start button, and Windows Vista are registered trademarks or trademarks of Microsoft Corporation. Microsoft product screen shots are reprinted with permission from Microsoft Corporation.

All other trademarks, service marks, and company names in this document or web site are properties of their respective owners.

Table of Contents

Preface	4
About this document.....	4
Document conventions.....	4
Intended audience	4
Revision History	4
Accessing product documentation	5
Getting Help.....	5
Comments	5
Introduction	6
Purpose	6
Linux OS NVMe/TCP	6
NVMe/TCP Connectivity and Multipathing.....	6
Connectivity Diagram	7
Hardware Requirements	8
Software requirements	8
Configuring VSP One Block 20 NVM Subsystem	9
Storage Port Settings	10
Configuring Topology Settings	10
Configuring the NVM Subsystem	11
Updating Namespace Security Settings.....	12
Creating a Namespace.....	12
Configuring Namespace Path Settings	13
Deleting or Modifying the Storage Configuration	14
Configuring NVMe/TCP on a Linux Host for VSP One Block 20 Components.....	15
Verifying that NVMe-CLI and DM-Multipath Packages are Installed.....	15
Configuring an NVMe/TCP Host	16
NVMe/TCP Target Discovery and Target Connect from Linux Host.....	17
Host NQN	17
NVME Target Discovery and Target Connect.....	17
Configuring Multipathing in RHEL and SLES	20
Configuring NVMe Native Multipath in RHEL 9.2	20
Configuring NVMe Native Multipath in SLES 15 SP5.....	22

Preface

About this document

This document shows how to configure NVMe/TCP with NVMe native multipathing for Hitachi Virtual Storage Platform One Block 20 (VSP One B20) NVM Subsystems in a Linux environment (Red Hat Enterprise Linux [RHEL] or SUSE Linux Enterprise Server [SLES]).

The storage system was configured using the Hitachi Command Control Interface (CCI). Linux configurations can be applied to RHEL 9.2 and SLES 15 SP5 Operating System (OS) environments; however, the procedures also apply to other RHEL, Oracle Linux, and SLES OS versions with minimal or no changes. For more information, see the OS vendor manuals.

Document conventions

This document uses the following typographic convention:

Convention	Description
Bold	- Indicates text in a window, including window titles, menus, menu options, buttons, fields, and labels. Example: Click OK. - Indicates emphasized words in list items.
<i>Italic</i>	Indicates a document title or emphasized words in text.
Monospace	Indicates text that is displayed on screen or entered by the user. Example: <code>pairdisplay -g oradb</code>

Intended audience

This document is intended for technical audience who need an overview of how to configure NVMe/TCP on an RHEL or SLES system with VSP One Block 20 family storage systems. To use this document, you must have a basic understanding of Linux, network configuration, Hitachi Command Control Interface, and VSP One Block 20 storage systems.

Revision History

Revision	Changes	Date
v1.0	Initial Release	July 2024

Accessing product documentation

Product user documentation is available on the Hitachi Vantara Support Site: <https://docs.hitachivantara.com/>. Check this site for the most current documentation, including important updates that may have been made after the release of the product. For more information on the operating system, see the operating system documentation. For the product support matrix, see the Product Compatibility Guide: <https://compatibility.hitachivantara.com/>

For respective OS support, check their support site:
RHEL: <https://access.redhat.com/products/red-hat-enterprise-linux>
SLES: <https://documentation.suse.com/>

Getting Help

[Hitachi Vantara Support Connect](#) is the destination for technical support of products and solutions sold by Hitachi Vantara. To contact technical support, log on to Hitachi Vantara Support Connect for contact information: https://support.hitachivantara.com/en_us/contact-us.html.

[Hitachi Vantara Community](#) is a global online community for customers, partners, independent software vendors, employees, and prospects. It is the destination to get answers, discover insights, and make connections. **Join the conversation today!** Go to community.hitachivantara.com, register, and complete your profile.

Comments

Please send us your comments on this document to GPSE-Docs-Feedback@hitachivantara.com. Include the document title and number, including the revision level (for example, -07), and refer to specific sections and paragraphs whenever possible. All comments become the property of Hitachi Vantara LLC.

Introduction

Hitachi Virtual Storage Platform One Block 20 (VSP One B20) is a new 2U all-NVMe storage appliance with three dedicated models. All models have the same capacity (72 NVMe flash drives, the appliance, + 2x media trays) and support FC, iSCSI, and NVMe TCP connectivity. The new capabilities remove complexity, such as data reduction always being on, Dynamic Drive Protection (DDP) removing complicated RAID setups, and Dynamic Carbon Reduction delivering real-world reductions in power consumption.

- VSP One Block 24 – 256GB Cache + SW Advanced Data Reduction (ADR) + 24 cores
- VSP One Block 26 – 768GB Cache + 2x Compression Accelerator Module (CAM) + 24 cores
- VSP One Block 28 – 1TB Cache + 4x CAM + 64 cores

This document describes the NVMe/TCP with NVMe native multipathing configuration for VSP One Block 20 storage NVM Subsystems in a Linux environment (Red Hat Enterprise Linux [RHEL] or SUSE Linux Enterprise Server [SLES]).

Purpose

This document describes the procedure for configuring NVMe/TCP across different OS environments, such as RHEL and SLES. In addition, this document emphasizes the configuration process of RHEL 9.2 and SLES 15 SP5, which covers setting up VSP One Block 20 with NVMe/TCP interfaces.

The purpose of this guide is to provide an overview of the implementation that includes VSP One Block 20 with NVMe/TCP port configuration, subsystem ID creation, NVMe namespace provisioning, host-side configuration for storage namespace detection, and native NVMe multipathing configuration.

Linux OS NVMe/TCP

NVMe/TCP is supported by most Linux operating systems, including Red Hat Enterprise Linux, Oracle Linux, and SUSE Linux Enterprise Servers.

Before you begin, see the following for more information:

- For NVMe/TCP support specifics, see the OS release notes, related installation documents, and storage configuration guides.
- For additional details about NVMe/TCP support, see the NIC user guides and configuration details for more information about NVMe/TCP support.

NVMe/TCP Connectivity and Multipathing

To connect NVMe/TCP, install a supported dual-port 100G NIC on a server running Linux x86-64 OS. Both NIC ports connect to VSP One Block 20 ports through two 100G Network Switches.

Using multipathing, a server communicates with a storage LUN through multiple physical paths between the Host NIC in the server and the storage controller devices. NVMe/TCP has a built-in native NVMe multipathing feature which handles multipathing for all NVMe namespaces.

Connectivity Diagram

The following diagram shows Port 3A and Port 4A of a VSP One Block 28 storage system connected through two 100G network switches:

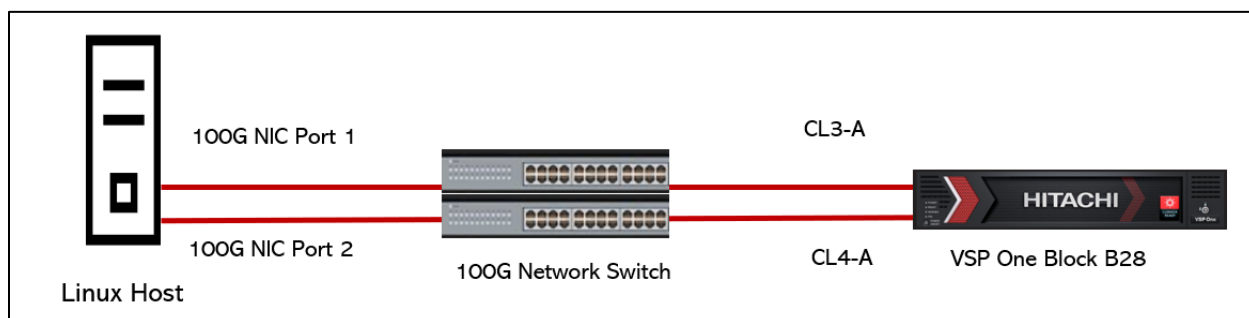


Figure 1: Port 3A and Port 4A of a VSP One Block 28 storage system connected through two 100G network switches

The following diagram shows the workflow of NVMe/TCP where the host NIC communicates with the NVM Subsystem to access the namespaces created from VSP One Block 20.

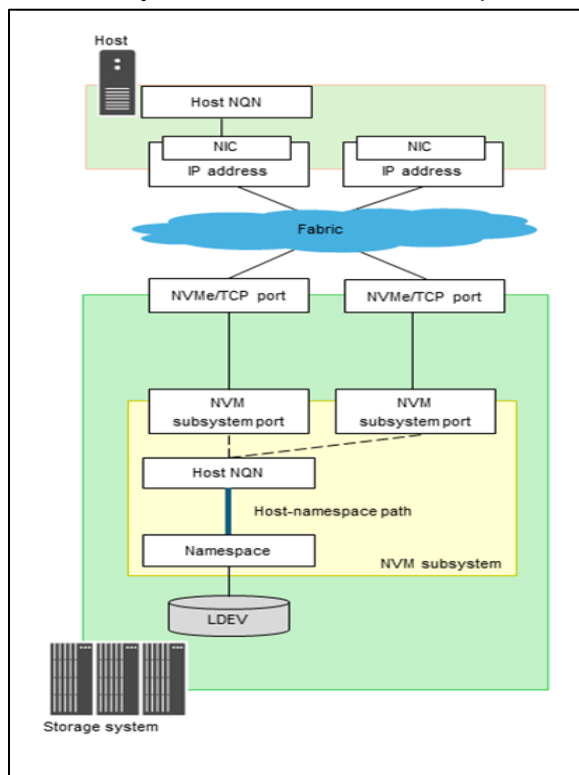


Figure 2: Workflow of NVMe/TCP

Hardware Requirements

The following lists the hardware used in this setup:

- NVMe/TCP host system: Hitachi Advanced Server HA820 G3
- Host NIC: Intel E810-CQDA2 100Gb/s Dual Port QSFP28
- CCI Server: HPE ProLiant DL380 Gen 10
- Switches: Arista 7280SR2A-48YC6, Mellanox SN2700
- Storage: Hitachi VSP One Block 28

Software Requirements

The following lists the software used in this setup:

- OS install media: RHEL 9.2, SLES 15 SP5
- CCI Software: Command Control Interface version 01-76-02/02

Configuring the VSP One Block 20 NVM Subsystem

This section describes how to configure the VSP One Block 20 NVM Subsystem with CCI.

Before you begin:

Confirm the storage system ESM IP address of the controllers and configure the 'horcm' file on a CCI management server to start the 'horcm' instance. For detailed information on CCI installation, configuration, and commands, see the Command Control Interface manuals. CCI server must be running with version 01-75-03/03 or later.

For this implementation, we created a CCI out-of-band configuration using a virtual command device (IPCMD) with controller ESM IP addresses.

Syntax of IPCMD under the HORCM_CMD section in the configuration definition file:

```
\\.\IPCMD-<GUM/ESM IP address>-<UDP communication port number>[-Unit ID]
```

- <GUM/ESM IP address>: Specifies an ESM IPv4 address of VSP One Block 20 Controller.
- <UDP communication port number>: Sets the UDP communication port number. The values (31001 and 31002) are fixed.
- [-Unit ID]: Sets the unit ID of the storage system for a multi-unit connection configuration. This can be omitted. For this implementation, we used only one storage unit, and it is omitted.

The following example shows a CCI configuration file used on a Linux System with Command Control Interface software. In the example, the ESM IP address of the two storage controllers are 172.23.68.12 and 172.23.68.13.

```
# cat horcm6811.conf
HORCM_MON
#ip address      service      poll(10ms)  timeout(10ms)
127.0.0.1        horcm6811    1000        3000

HORCM_CMD
#dev_name dev_name dev_name dev_name
\\.\IPCMD-172.23.68.12-31001 \\.\IPCMD-172.23.68.12-31002 \\.\IPCMD-
172.23.68.13-31001 \\.\IPCMD-172.23.68.13-31002

# cat /etc/services |grep horcm6811
horcm6811      485232/udp      #horcm6811

# horcmstart.sh 6811
HORCM inst 6811 has already been running.
```

Storage Port Settings

This section describes how to configure the 100G TCP storage port.

Verify NVMe mode settings for the storage ports (3A and 4A) by completing the following steps:

1. Verify that the port mode is 'NVME_TCP'.
2. Verify the port mode using the following CCI raidcom command:

```
# raidcom get port -key detail -I6811
PORT  TYPE ATTR  SPD LPID FAB CONN SSW  SL  Serial# WWN          PHY_PORT PORT_MODE RSGID SECURITY
CL3-A NVME_TCP TAR   100G  00 N  UNKN N   0  810008 -          -    NVME_TCP  0 -
```

By default, the port mode of 100G CHB is NVME_TCP because the default support protocol is NVMe/TCP.

Configuring Topology Settings

Configure the 100G TCP storage ports by completing the following steps:

1. Assign network IP addresses to the 100G TCP storage ports in different subnet ranges, as shown in the following example.

In the following example, network IP addresses, 192.168.10.30 and 192.168.20.30, are assigned against CL3-A and CL4-A 100G storage ports.

```
# raidcom modify port -port CL3-A -ipv4_address 192.168.10.30
-ipv4_subnetmask 255.255.255.0 -I6811
```

```
# raidcom modify port -port CL4-A -ipv4_address 192.168.20.30
-ipv4_subnetmask 255.255.255.0 -I6811
```

2. Assign the network IP addresses to the 100G TCP storage ports and host NIC ports in the same range of IP addresses in two different subnets.
3. Verify the IP address configuration by running the following command:

```
# raidcom get port -port CL3-A -key opt -IH6811
PORT : CL3-A
TCP_OPT : IPV6_D : SACK_E : DACK_E : INS_D : VTAG_D
TCP_MTU : 1500
WSZ : 1024KB
KA_TIMER : 60
TCP_PORT_IO_CONTROLLER : 4420
TCP_PORT_DISCOVERY_CONTROLLER : 8009
IPV4_ADDR : 192.168.10.30
IPV4_SMSK : 255.255.255.0
IPV4_GWAD : 0.0.0.0
IPV6_ADDR_INF : INV : AM : fe80::
IPV6_GADR_INF : INV : AM : ::
IPV6_GWAD_INF : INV : :: : ::
VLAN_ID : -
```

```

VP_MODE : D
IPV6_GADR2_INF : INV : AM : ::
MAC_ADDR : f8:48:97:47:e4:0d

# raidcom get port -port CL4-A -key opt -IH6811
PORT : CL4-A
TCP_OPT : IPV6_D : SACK_E : DACK_E : INS_D : VTAG_D
TCP_MTU : 1500
WSZ : 1024KB
KA_TIMER : 60
TCP_PORT_IO_CONTROLLER : 4420
TCP_PORT_DISCOVERY_CONTROLLER : 8009
IPV4_ADDR : 192.168.20.30
IPV4_SMSK : 255.255.255.0
IPV4_GWAD : 0.0.0.0
IPV6_ADDR_INF : INV : AM : fe80::
IPV6_GADR_INF : INV : AM : ::
IPV6_GWAD_INF : INV : :: : ::
VLAN_ID : -
VP_MODE : D
IPV6_GADR2_INF : INV : AM : ::
MAC_ADDR : f8:48:97:47:e6:01

```

Configuring the NVM Subsystem

Create an NVM Subsystem and register the 100G TCP storage ports to the NVM Subsystem by completing the following steps:

1. Create an NVM Subsystem on the storage system with the CCI raidcom command.
2. Verify whether there is an existing NVM Subsystem and then create the NVM Subsystem with an available ID.

In the following example, the NVM Subsystem is created with ID 1:

```

# raidcom add nvm_subsystem -nvm_subsystem_id 1 -nvm_subsystem_name TCP_NVMe_RHEL9u2
-request_id auto -I6811

# raidcom get nvm_subsystem -nvm_subsystem_id 1 -IH6811
NVMSS_ID  RGID  NVMSS_NAME      SECURITY T10PI    HMD          HMO_BITS  V_NVMSS_ID
1          0    TCP_NVMe_RHEL9u2      ENABLE   DISABLE   LINUX/IRIX   -          -

# raidcom get nvm_subsystem -key opt -IH6811
NVMSS_ID  NVMSS_NAME                                NVMSS_NQN
1    TCP_NVMe_RHEL9u2  nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10008-
nvmssid.00001

```

3. Register the 100G TCP storage ports with the NVM subsystem using the CCI raidcom command.

In the following example, Storage Ports CL3-A and CL4-A ports are registered with NVM Subsystem ID 1:

```

# raidcom add nvm_subsystem_port -nvm_subsystem_id 1 -port CL3-A -request_id auto -IH6811
# raidcom add nvm_subsystem_port -nvm_subsystem_id 1 -port CL4-A -request_id auto -IH6811

# raidcom get nvm_subsystem_port -nvm_subsystem_id 1 -IH6811

```

PORT	NVMSS_ID	NVMSS_NAME
CL3-A	1	TCP_NVMe_RHEL9u2
CL4-A	1	TCP_NVMe_RHEL9u2

Updating Namespace Security Settings

Update the Namespace security settings by completing the following steps:

1. Make a note of the Host NQN to use when configuring the NVM Subsystem. To check the Host NQN on a Linux system, run the following command:

```
# cat /etc/nvme/hostnqn
nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

2. Set the Host NQN to allow access to the NVM Subsystem with the CCI raidcom command, as shown in the following example:

```
# raidcom add host_nqn -nvme_subsystem_id 1 -host_nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843 -request_id auto -I6811

# raidcom get host_nqn -nvme_subsystem_id 1 -I6811
NVMSS_ID  NVMSS_NAME                                HOST_NQN
1  TCP_NVMe_RHEL9u2  nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

Creating a Namespace

Create a Namespace by registering the LDEV with the NVM Subsystem. A Namespace is then created for that LDEV with a unique Namespace ID.

1. Verify whether the LDEV is free to assign. If required, create new LDEVs.

```
# raidcom get ldev -ldev_id 12:00 -IH6811

Serial#   : 810008
LDEV      : 4608
SL        : -
CL        : -
VOL_TYPE  : NOT DEFINED
SSID      : 0016
RSGID     : 0

# raidcom get ldev -ldev_id 12:01 -IH6811

Serial#   : 810008
LDEV      : 4609
SL        : -
CL        : -
VOL_TYPE  : NOT DEFINED
SSID      : 0016
RSGID     : 0
```

2. Create new LDEVs with the CCI raidcom command, as shown in the following example:

```
# raidcom add ldev -parity_grp_id 1-1 -ldev_id 12:00 -capacity 10G -I6811
# raidcom add ldev -parity_grp_id 1-1 -ldev_id 12:01 -capacity 10G -I6811
```

3. Format the newly created LDEVs using the raidcom command, as shown in the following example:

```
# raidcom initialize ldev -ldev_id 12:00 -operation fmt -I6811
# raidcom initialize ldev -ldev_id 12:01 -operation fmt -I6811
```

4. Create the Namespace for the LDEVs by registering them with the Storage Subsystem ID, as shown in the following example:

```
# raidcom add namespace -nvm_subsystem_id 1 -ns_id auto -ldev_id 12:00 -request_id auto -I6811
# raidcom add namespace -nvm_subsystem_id 1 -ns_id auto -ldev_id 12:01 -request_id auto -I6811
```

5. Verify that the Namespaces are created for the LDEVs in the storage subsystem:

```
# raidcom get namespace -nvm_subsystem_id 1 -IH6811

1  PegMR2U_TCP_NVMe_RHEL9u2_PB1  1  4608  20971520
1  PegMR2U_TCP_NVMe_RHEL9u2_PB1  2  4609  20971520
```

Configuring Namespace Path Settings

To configure the Namespace path settings, complete the following steps:

1. Register the Host NQN to the Namespace ID to allow access for the Host by setting the Host NQN-Namespace Path with the CCI Namespace Path configuration command, as shown in the following example:

```
# raidcom add namespace_path -nvm_subsystem_id 1 -ns_id 1 -host_nqn nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843 -request_id auto -I6811

# raidcom add namespace_path -nvm_subsystem_id 1 -ns_id 2 -host_nqn nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843 -request_id auto -I6811
```

2. Verify the Namespace path configuration by running the raidcom command, as shown in the following example:

```
# raidcom get namespace_path -nvm_subsystem_id 1 -IH6811

1  PegMR2U_TCP_NVMe_RHEL9u2_PB1  1  4608  nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
1  PegMR2U_TCP_NVMe_RHEL9u2_PB1  2  4609  nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

Deleting or Modifying the Storage Configuration

Before you delete or modify the storage configuration, ensure that there is no I/O running on your storage devices. Additionally, unmount all storage devices before starting any storage configuration changes.

The following are examples of commands you can use to change the settings:

- To delete the Namespace path for an LDEV:

```
# raidcom delete namespace_path -nvm_subsystem_id 1 -ns_id 1 -host_nqn nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843 -request_id auto -I6811
```

- To delete the Namespace for an LDEV:

```
# raidcom delete namespace -nvm_subsystem_id 1 -ns_id 1 -request_id auto -I6811
```

- To delete the Host NQN from the NVM Subsystem:

```
# raidcom delete host_nqn -host_nqn nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843 -nvm_subsystem_id 1 -request_id auto -I6811
```

- To delete the NVM Subsystem Port from the NVM Subsystem:

```
# raidcom delete nvm_subsystem_port -nvm_subsystem_id 1 -port CL3-A -request_id auto -I6811
```

- To delete the NVM Subsystem:

```
# raidcom delete nvm_subsystem -nvm_subsystem_id 1 -request_id auto -I6811
```

- To display the Host NQN login status for CL3-A:

```
# raidcom get port -port CL3-A -key login_host_nqn -I6811
```

```
PORT      LOGIN_STATUS  HOST_NQN
CL3-A     LOGIN         nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

Configuring NVMe/TCP on a Linux Host for VSP One Block 20 Components

This section describes how to install and configure the NVMe package on RHEL and SLES. In addition, NVMe target discovery and connection is described with examples.

Before you begin

- Install the OS by following the instructions in the vendor installation documents.
- Verify that the NVMe/TCP NICs are detected by the OS inbox driver. If required, install the latest NVMe/TCP supported driver provided by the NIC product vendor.

Verifying that NVMe-CLI and DM-Multipath Packages are Installed

This section provides examples for verifying whether the NVMe-CLI and DM-Multipath packages are installed based on your OS version:

1. Verify whether NVMe-CLI and DM-Multipath are installed, as shown in the following examples:

On a RHEL 9.2 system:

```
# rpm -qa | grep nvme-cli
nvme-cli-2.2.1-2.el9.x86_64

# rpm -qa | grep multipath
device-mapper-multipath-libs-0.8.7-20.el9.x86_64
device-mapper-multipath-0.8.7-20.el9.x86_64
```

On a SLES 15 SP5 system:

```
# rpm -qa | grep nvme-cli
nvme-cli-bash-completion-2.2.1+3.gd028407-150500.2.4.noarch
nvme-cli-2.2.1+3.gd028407-150500.2.4.x86_64

# rpm -qa | grep multipath
multipath-tools-0.9.4+71+suse.c648a77-150500.1.2.x86_64
```

2. If the NVMe-CLI package is not installed during the OS installation, you can install the package manually from the OS installation media or from the local repository for RHEL and SLES:

Example: Install nvme-cli package in RHEL:

```
# yum install nvme-cli
```

Example: Install nvme-cli package in SLES:

```
# zypper install nvme-cli
```

Configuring an NVMe/TCP Host

This section shows how to configure the NIC network on the RHEL and SLES host so it can communicate with the target VSP One Block 20 storage system.

1. Verify that the `hostnqn` is generated by running the following command:

```
# cat /etc/nvme/hostnqn
nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

2. Configure static private IP addresses in different subnet ranges on the ethernet interfaces of the installed NIC on the host by following the respective OS documentation.
3. Verify the static IP addresses assigned on both the ethernet interfaces are working properly on the ethernet interfaces by running the following `'ip addr show'` command:

```
# ip addr show
```

```
.....

ens2f0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP
group default qlen 1000
link/ether 30:3e:a7:00:35:0e brd ff:ff:ff:ff:ff:ff
altname enp35s0f0
inet 192.168.10.45/24 brd 192.168.10.255 scope global noprefixroute
    ens2f0
    valid_lft forever preferred_lft forever
inet6 fe80::323e:a7ff:fe00:350e/64 scope link noprefixroute
    valid_lft forever preferred_lft forever

ens2f1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP
group default qlen 1000
link/ether 30:3e:a7:00:35:0f brd ff:ff:ff:ff:ff:ff
altname enp35s0f1
inet 192.168.20.45/24 brd 192.168.20.255 scope global noprefixroute
    ens2f1
    valid_lft forever preferred_lft forever
inet6 fe80::323e:a7ff:fe00:350f/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
.....
```

4. Perform a network connectivity check to ensure that the target IP addresses are pinging successfully from the host without any packet loss:

```
# ping 192.168.10.30
PING 192.168.10.30 (192.168.10.30) 56(84) bytes of data.
64 bytes from 192.168.10.30: icmp_seq=1 ttl=64 time=0.424 ms
64 bytes from 192.168.10.30: icmp_seq=2 ttl=64 time=0.286 ms
64 bytes from 192.168.10.30: icmp_seq=3 ttl=64 time=0.254 ms
64 bytes from 192.168.10.30: icmp_seq=4 ttl=64 time=0.522 ms
--- 192.168.10.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3104ms
rtt min/avg/max/mdev = 0.254/0.371/0.522/0.107 ms
```

NVMe/TCP Target Discovery and Connect from Linux Host

This section describes how to configure the NVMe target discovery and connect from the Linux host.

Before you start NVMe target discovery, verify the following using the procedures provided in previous sections of this document:

- NVMe-CLI package is installed on the Linux Host.
- Physical network connectivity is established between the host NIC Port and the storage port through the Network Switch.
- Network IP addresses are assigned to the NIC ports on the host.
- VSP One Block 20 is configured and the namespaces created for the LDEVs in the Storage NVM Subsystem and Namespace path are defined with the Host NQN.

Host NQN

Before starting the discovery process, verify the Host NQN as shown in the following example:

```
# cat /etc/nvme/hostnqn
nqn.2014-08.org.nvmexpress:uuid:37333550-3035-4753-4833-333053573843
```

NVMe Target Discovery and Target Connect

1. Load the 'nvme_tcp' module (if not already loaded):

```
# modprobe nvme_tcp
```

2. Discover the NVMe controllers by running the following NVMe discovery command:

Command syntax:

```
nvme discover --transport=tcp --traddr=<Target Controller IP> --host-traddr=<Host NIC IP address> --trsvcid=8009
```

Port 8009 is the default TCP port for NVMe/TCP discovery controllers.

Example: NVMe/TCP target discovery

```
# nvme discover --transport=tcp --traddr=192.168.10.30 --host-traddr=192.168.10.45 --trsvcid=8009
```

```
Discovery Log Number of Records 1, Generation counter 5
=====Discovery Log Entry 0=====
trtype: tcp
adrfam: ipv4
subtype: nvme subsystem
treq: not specified, sq flow control disable supported
portid: 0
```

```
trsvcid: 4420
subnqn: nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10008-
nvmsid.00001
traddr: 192.168.10.30
eflags: not specified
sectype: none
```

```
# nvme discover --transport=tcp --traddr=192.168.20.30 --host-
traddr=192.168.20.45 --trsvcid=8009
```

```
Discovery Log Number of Records 1, Generation counter 5
====Discovery Log Entry 0=====
trtype: tcp
adrfam: ipv4
subtype: nvme subsystem
treq: not specified, sq flow control disable supported
portid: 1
trsvcid: 4420
subnqn: nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10008-
nvmsid.00001
traddr: 192.168.20.30
eflags: not specified
sectype: none
```

3. Update the `/etc/nvme/discovery.conf` file by adding the parameters used for discovering the controllers.

In the event of a path failure, the NVM subsystem tries to reconnect to the failed controller for a specific amount of time, which is managed by the 'ctrl-loss-tmo' option of the `nvme connect` command. The default value of 'ctrl-loss-tmo' is 600 seconds, and beyond this time, the failed path does not connect automatically. Set the 'ctrl-loss-tmo' option to -1 so that the NVM Subsystem keeps trying to reconnect for an infinite period.

To set the value of 'ctrl-loss-tmo' to infinite, modify the `/etc/nvme/discovery.conf` file by setting the value to -1, as shown in the following example:

```
# cat /etc/nvme/discovery.conf

# Used for extracting default parameters for discovery
#
# Example:
# --transport=<trtype> --traddr=<traddr> --trsvcid=<trsvcid> --host-traddr=<host-traddr>
# --host-iface=<host-iface>
--transport=tcp --traddr=192.168.10.30 --host-traddr=192.168.10.45 --trsvcid=8009
--ctrl-loss-tmo=-1
--transport=tcp --traddr=192.168.20.30 --host-traddr=192.168.20.45 --trsvcid=8009
--ctrl-loss-tmo=-1
```

4. Connect to all the discovered controllers by running the following command:

```
# nvme connect-all
```

5. Enable the `nvme-autoconnect` service so that all the controllers are connected automatically during bootup by running the following command:

```
# systemctl enable nvme-autoconnect.service
# systemctl start nvme-autoconnect.service
```

6. List all the NVMe namespaces that are connected to the Linux host, as shown in the following example:

```
# nvme list
```

Node FW Rev	Generic	SN	Model	Namespace	Usage	Format
/dev/nvme1n1 0 B A3020140	/dev/ng1n1	8-10008-00001	HITACHI SVOS-RF-System	1	10.74 GB / 10.74 GB	512 B +
/dev/nvme1n2 0 B A3020140	/dev/ng1n2	8-10008-00001	HITACHI SVOS-RF-System	2	10.74 GB / 10.74 GB	512 B +

7. List the available paths connected to the NVM Subsystem:

```
# nvme list-subsys /dev/nvme1n1
```

```
nvme-subsys1 - NQN=nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10008-
nvmsid.00001
\
+- nvme1 tcp traddr=192.168.20.30,trsvcid=4420,host_traddr=192.168.20.45 live optimized
+- nvme2 tcp traddr=192.168.10.30,trsvcid=4420,host_traddr=192.168.10.45 live optimized
```

Configuring Multipathing in RHEL and SLES

This section describes how to configure NVMe native multipathing for VSP One Block 20 NVMe namespaces on RHEL 9.2 and SLES 15 SP5 systems.

Configuring NVMe Native Multipath in RHEL 9.2

NVMe native multipathing is enabled by default in RHEL 9.2. If it is not enabled, complete the following steps:

1. To enable NVME multipathing, set 'nvme_core.multipath=Y' for the 'nvme-core' module 'multipath' option to 'Y'.
2. Verify the currently running kernel and identify the 'vmlinuz' kernel file in the /boot directory as shown in the following example. Then, run the grubby command to enable Native NVMe multipathing:

```
# uname -r
5.14.0-284.11.1.el9_2.x86_64

# cd /boot
# ls -l
total 250620
-rw-r--r--. 1 root root      215622 Apr 12  2023 config-5.14.0-284.11.1.el9_2.x86_64
drwx----- 3 root root      4096 Jan  1  1970 efi
drwxr-xr-x. 2 root root      4096 Mar 22 15:15 extlinux
drwx----- 3 root root        50 Mar 22 15:22 grub2
-rw----- 1 root root 134140228 Mar 22 15:20 initramfs-0-rescue-
a52e08acd3984810a7584c0a6c7f1ef7.img
-rw----- 1 root root  58436642 Apr 30 17:32 initramfs-5.14.0-284.11.1.el9_2.x86_64.img
-rw----- 1 root root  33546752 May  2 07:06 initramfs-5.14.0-
284.11.1.el9_2.x86_64kdump.img
drwxr-xr-x. 3 root root        21 Mar 22 15:19 loader
lrwxrwxrwx. 1 root root        52 Mar 22 15:19 symvers-5.14.0-284.11.1.el9_2.x86_64.gz ->
/lib/modules/5.14.0-284.11.1.el9_2.x86_64/symvers.gz
-rw----- 1 root root   5921749 Apr 12  2023 System.map-5.14.0-284.11.1.el9_2.x86_64
-rwxr-xr-x. 1 root root  12176920 Mar 22 15:20 vmlinuz-0-rescue-
a52e08acd3984810a7584c0a6c7f1ef7
-rwxr-xr-x. 1 root root  12176920 Apr 12  2023 vmlinuz-5.14.0-284.11.1.el9_2.x86_64

# grubby --args=nvme_core.multipath=Y --update-kernel /boot/vmlinuz-5.14.0-
284.11.1.el9_2.x86_64
```

3. Verify that DM multipath is disabled and remove any /etc/multipath.conf files by running the following commands:

```
# systemctl disable multipathd
# systemctl stop multipathd
# systemctl status multipathd
# multipath -F
# rm /etc/multipath.conf
```

4. Recreate the 'initramfs' file and restart the system by running the following commands:

```
# dracut -f -v
# reboot
```

- Verify that Native NVMe multipath is enabled. If the output is 'Y', Native NVMe multipath is enabled.

```
# cat /sys/module/nvme_core/parameters/multipath
Y
```

- Create a udev rule for the VSP One Block 20 NVM Subsystem to enable round-robin load balancing for NVMe Native multipath on RHEL 9.2. Create a 71-nvme-iopolicy-hitachi-SVOS.rules file under the /lib/udev/rules.d directory:

```
# cd /lib/udev/rules.d
# pwd
/lib/udev/rules.d

# cat /lib/udev/rules.d/71-nvme-iopolicy-hitachi-SVOS.rules
# Enable round-robin for hitachi SVOS
ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{model}=="HITACHI SVOS-RF-System",
ATTR{iopolicy}="round-robin"
```

- Restart the system.

```
# reboot
```

- Verify all NVMe devices and namespaces on the host. The 'nvme list' command only displays a single NVMe device for an NVMe disk with NVMe Native multipath.

```
# nvme list
```

Node Format	Generic FW Rev	SN	Model	Namespace Usage	
/dev/nvme1n1 512 B + 0 B	/dev/ng1n1 A3020140	8-10008-00001	HITACHI SVOS-RF-System	1	10.74 GB / 10.74 GB
/dev/nvme1n2 512 B + 0 B	/dev/ng1n2 A3020140	8-10008-00001	HITACHI SVOS-RF-System	2	10.74 GB / 10.74 GB

The 'nvme list-subsys' command displays NVM subsystems with path information, as shown in the following example:

```
# nvme list-subsys /dev/nvme1n1

nvme-subsys1 - NQN=nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10008-nvmssid.00001
\
+- nvme1 tcp traddr=192.168.20.30,trsvcid=4420,host_traddr=192.168.20.45 live optimized
+- nvme2 tcp traddr=192.168.10.30,trsvcid=4420,host_traddr=192.168.10.45 live optimized
```

- Verify that the VSP One Block 20 NVM Subsystem is detected. Additionally, verify the effective IO Policy for the VSP One Block 20 NVM Subsystem by running the following commands:

```
# cat /sys/class/nvme-subsystem/nvme-subsys*/model
HITACHI SVOS-RF-System

# cat /sys/class/nvme-subsystem/nvme-subsys*/iopolicy
round-robin
```

- Create a partition and a filesystem on the NVMe Native multipath-enabled disk.

You can create a partition and a filesystem using NVMe/TCP devices in the `/dev/` directory:

```
# nvme list
Node                               Generic      SN          Model
Namespace Usage                    Format      FW Rev
-----
/dev/nvme1n1                      /dev/ng1n1  8-10008-00001  HITACHI SVOS-RF-System  1  10.74 GB / 10.74 GB
512 B + 0 B A3020140
/dev/nvme1n2                      /dev/ng1n2  8-10008-00001  HITACHI SVOS-RF-System  2  10.74 GB / 10.74 GB
512 B + 0 B A3020140

# fdisk /dev/nvme1n1
# mkfs.xfs /dev/nvme1n1p1
```

11. Obtain the UUID of the file systems and update the `/etc/fstab` file for mounting the filesystem during boot:

```
# blkid
.....

/dev/nvme1n1p1: UUID="10125fac-4d94-467a-81df-39ad9e630d9d" TYPE="xfs" PARTUUID="e9e45905-01"
/dev/nvme1n2p1: UUID="ccc156c3-5dad-4932-a49e-45d3538efb0f" TYPE="xfs" PARTUUID="2a81a1b6-01"
.....

# cat /etc/fstab
UUID=10125fac-4d94-467a-81df-39ad9e630d9d /fs0 xfs defaults,_netdev 0 0
UUID=ccc156c3-5dad-4932-a49e-45d3538efb0f /fs1 xfs defaults,_netdev 0 0

# mount -a
# df -h|grep nvme
/dev/nvme1n1p1 10G 1.1G 9.0G 11% /fs0
/dev/nvme1n2p1 10G 1.1G 9.0G 11% /fs1
```

Configuring NVMe Native Multipath in SLES 15 SP5

NVMe Native multipath is enabled by default in SLES 15 SP5. If it is not enabled by default, set the 'nvme-core' module option to 'Y' by completing the following steps:

1. If NVMe native multipathing is disabled, enable it by running the following commands:

```
# echo "options nvme_core multipath=Y" > /etc/modprobe.d/50-nvme_core.conf
# dracut -f -v
```

2. Restart the system.

```
# reboot
```

3. Verify that Native NVMe multipath is enabled. If the output is 'Y', Native NVMe multipath is successfully enabled.

```
# cat /sys/module/nvme_core/parameters/multipath
Y
```

4. Create a udev rule for the VSP One Block 20 NVM Subsystem to enable round-robin load balancing for NVMe multipath. Create a '71-nvme-iopolicy-hitachi-SVOS.rules' file in the '/lib/udev/rules.d' directory:

```
# cd /usr/lib/udev/rules.d
# pwd
/usr/lib/udev/rules.d

# cat /usr/lib/udev/rules.d/71-nvme-iopolicy-hitachi-SVOS.rules
# Enable round-robin for hitachi SVOS
ACTION=="add", SUBSYSTEM=="nvme-subsystem", ATTR{model}=="HITACHI SVOS-RF-System",
ATTR{iopolicy}="round-robin"
```

5. Restart the system.

```
# reboot
```

6. Verify all NVMe devices and namespaces on the host. The 'nvme list' command displays only a single device name for an NVMe disk with NVMe Native multipath.

```
# nvme list
```

Node Format	Generic FW Rev	SN	Model	Namespace	Usage
/dev/nvme2n1	/dev/ng2n1	8-10029-00001	HITACHI SVOS-RF-System	1	10.74 GB / 10.74 GB
512 B + 0 B	A3022040				
/dev/nvme2n2	/dev/ng2n2	8-10029-00001	HITACHI SVOS-RF-System	2	10.74 GB / 10.74 GB
512 B + 0 B	A3022040				

The 'nvme list-subsys' command displays NVM Subsystems with path information, as shown in the following example:

```
# nvme list-subsys /dev/nvme2n1
nvme-subsys2 - NQN=nqn.1994-04.jp.co.hitachi:nvme:storage-subsystem-sn.8-10029-nvmssid.00001
\
+- nvme2 tcp traddr=192.168.10.30,trsvcid=4420,host_traddr=192.168.10.45 live optimized
+- nvme3 tcp traddr=192.168.20.30,trsvcid=4420,host_traddr=192.168.20.45 live optimized
```

7. Verify that the VSP One Block 20 NVM Subsystem is detected. Additionally, verify the effective IO Policy for the VSP One Block 20 NVM Subsystem by running the following commands:

```
# cat /sys/class/nvme-subsystem/nvme-subsys*/model
HITACHI SVOS-RF-System

# cat /sys/class/nvme-subsystem/nvme-subsys*/iopolicy
round-robin
```

8. Create a partition and make a filesystem on the NVMe Native multipath disk.

You can create a partition and make a filesystem using NVMe/TCP devices in the /dev/ directory:

```
# fdisk /dev/nvme2n1
# mkfs.xfs /dev/nvme2n1p1
```

9. Obtain the UUID of the file systems and update the /etc/fstab file for mounting the filesystem during boot:

```
# blkid
.....
/dev/nvme2n1p1: UUID="e4b7266f-f890-40f3-83ec-64a905acc98d" BLOCK_SIZE="512" TYPE="xfs"
PARTUUID="1a55ef0c-4c1f-4475-ab5d-27ac011c7951"
/dev/nvme2n2p1: UUID="f5059884-916e-40a4-a332-23cbf32dc523" BLOCK_SIZE="512" TYPE="xfs"
PARTUUID="ae3cb851-e1b4-4437-b37f-4017b6369d2c"
```

..... *

cat /etc/fstab

```
UUID=e4b7266f-f890-40f3-83ec-64a905acc98d    /fs0    xfs     defaults,_netdev    0 0
UUID=f5059884-916e-40a4-a332-23cbf32dc523    /fs1    xfs     defaults,_netdev    0 0
```

mount -a

df -h|grep nvme

```
/dev/nvme2n1p1    10G    1.1G    9.0G    11% /fs0
/dev/nvme2n2p1    10G    1.1G    9.0G    11% /fs1
```

Hitachi Vantara

Corporate Headquarters 2535 Augustine Drive
Santa Clara, CA 95054 USA www.HitachiVantara.com community.HitachiVantara.com

Regional Contact Information

Americas: +1 866 374 5822 or info@hitachivantara.com

Europe, Middle East and Africa: +44 (0) 1753 618000 or info.emea@hitachivantara.com

Asia Pacific: +852 3189 7900 or info.marketing.apac@hitachivantara.com

